

SOUNDARIYA VIJAYAKUMAR

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🔗 soundariya.com

PROFESSIONAL SUMMARY

Software Engineer with 3 years of production experience building scalable backend systems, RESTful APIs and distributed applications using Java, Spring Boot, and cloud technologies. Proven track record in microservices architecture, database optimization, and CI/CD automation with strong focus on reliability and scaling.

Education

Northeastern University

Master of Science in Computer Engineering *Gpa: 3.75*

Sep. 2024 – May 2026

Seattle, WA

Rajalakshmi Institute of Technology

Bachelor of Engineering in Electronics & Communication Engineering

Sep. 2016 – May 2020

Chennai, India

Relevant Coursework:

Operating Systems, Data Structures & Algorithms, Computer Architecture, Database, Distributed Systems, Machine Learning

Experience

LTIMindtree

Software Engineer

Jan 2021 – Dec 2023

Chennai, India

- Developed custom listing and filtering modules in Sitefinity CMS using **C#** and **SQL Server**, managing deployments and maintenance for 8 multilingual client websites
- Optimized database queries by 60% through strategic indexing and reduced **API response** times by 40% through code refactoring and caching strategies
- Contributed to **CI/CD** implementation with Azure DevOps, improving deployment reliability and reducing downtime by 50%.
- Participated in Agile workflows conducting code reviews, analyzing system bottlenecks, and implementing improvements to backend scalability and stability

Projects

Scalable Messaging Backend – Microservices Architecture | *Java, WebSockets, RabbitMQ, AWS EC2* **October 2025**

- Built distributed real-time chat system using microservices architecture with **Spring Boot RESTful APIs**, **RabbitMQ** message broker for async communication, and **WebSocket** servers for real-time messaging
- Developed scalable, fault-tolerant backend handling 10k+ concurrent connections using load balancing, **ElastiCache**, **Redis**, **DynamoDB** for persistent storage, and consistent hashing for data partitioning
- Conducted performance testing and benchmarking with **JMeter**, achieving **sub-100ms P95 latency**
- Implemented comprehensive monitoring using **Prometheus**, **Grafana**, and **CloudWatch** for metrics collection, real-time profiling, and resource optimization

Custom Shell for Linux — C++ — Systems Programming | *C++, System calls-Linux/Unix*

April 2025

- Developed Unix-like shell in C++ with multi-threaded process management, command parsing, piping, I/O redirection, and signal handling
- Debugged memory leaks and performance issues using Valgrind and GDB, ensuring efficient resource utilization

AI Powered Resume Analyzer & Job Matcher | *Python, BERT, Machine learning, PyTorch*

October 2025

- Built production-ready NLP system leveraging **BERT** transformer architecture (encoder layers, & token embeddings) for semantic skill extraction from unstructured resume text, achieving 92% F1-score on skill classification task
- Designed ML inference pipeline combining transformer-based embeddings with traditional NLP techniques (TF-IDF, cosine similarity), implementing batch processing and model optimization to deliver real-time predictions with 95% accuracy on job-resume matching

Technical Skills

Languages: C++ (C++11/14), Java, Python

Backend & APIs: Java, Spring Boot, REST APIs, Microservices Architecture

Databases: SQL (MySQL, Postgres), NoSQL (DynamoDB), Database Optimization, Redis

Testing: JUnit & JMeter

Cloud & AWS: AWS EC2, AWS S3, AWS CloudWatch, Containerization, Apache Kafka, AWS SQS

CI/CD & DevOps: Jenkins, Terraform, Azure DevOps, Docker, Git, Pipeline Management

Machine Learning: PyTorch, BERT/Transformers, Deep Learning, NLP, Model Optimization

Additional: C#, ASP.NET, RabbitMQ, WebSockets, Agile/Scrum